

# Preet Raval

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## PROFESSIONAL SUMMARY

DevOps and full-stack engineer who has been the sole engineer behind 5 production systems used daily by 50+ employees at American Circuits Inc., including traveler management, inventory, internal AI tooling, and the infrastructure supporting them. Owns systems end-to-end: schema design, REST APIs, UI, deployment, and ongoing uptime. Core stack is Python, FastAPI, Next.js, PostgreSQL, Docker, and Linux, with hands-on experience in CI/CD automation, LLM/OCR integration, and enterprise SSO/MFA security

## SKILLS

**Languages & Frameworks:** Python, TypeScript/JavaScript, Java, C#/.NET, React, Next.js, FastAPI, Java Spring Boot, Tailwind CSS, HTML5/CSS3, RESTful APIs, Microservices

**Databases:** PostgreSQL, MySQL, MongoDB, SQL Server, Oracle Database, SQL Optimization, Data Modeling, ORM, BigQuery

**DevOps & Cloud:** Docker, Kubernetes, Nginx, CI/CD, GitHub Actions, Linux, AWS, Azure, Vercel, Cloudflare Tunnels, On-Premise Infrastructure, Server Administration, Observability, ArgoCD, Terraform

**AI / ML:** LLM Integration (Claude API, ChatGPT API), Prompt Engineering, RAG, OCR, Agentic AI, MCP (Model Context Protocol), TensorFlow, PyTorch, HuggingFace Transformers, Pandas, NumPy

**Security & Networking:** SSO/MFA, Role-Based Access Control, Penetration Testing, Vulnerability Assessment, SIEM, TCP/IP, Subnetting, VLANs, Network Security, Encryption, IAM

**Other:** ERP/MRP/CRM (ODOO), Jira, Git, Agile/SDLC, Data Analytics, Tableau, Technical Documentation

## EDUCATION

**UNC Charlotte** B.A., Computer Science - Charlotte, NC Jan 2023 – Dec 2025

**Vellore Institute of Technology** B.Tech, Computer Science (Transfer) - India Sep 2020 – Dec 2022

## EXPERIENCE

**DevOps Engineer, AI Systems** Mar 2026 – Present  
American Circuits Inc. Charlotte, NC

- Owned and scaled Linux-based production infrastructure supporting business-critical applications across multiple environments.
- Architected and owned CI/CD pipelines using GitHub Actions, cutting deployment effort significantly and driving consistent, reliable releases.
- Led containerization and orchestration of microservices using Docker and Kubernetes, achieving zero-downtime deployments and improving overall system reliability.
- Owned security posture end-to-end: implemented RBAC, secrets management, and infrastructure hardening aligned with security best practices.
- Drove automation of reporting, operational workflows, and third-party integrations using Python and REST APIs, eliminating multiple hours of manual work per week.
- Took full ownership of backups, monitoring, system patching, and incident response, maintaining 99%+ uptime across production systems.

**Software Engineer** Jun 2025 – Mar 2026  
American Circuits Inc. Charlotte, NC

- Architected and shipped enterprise-grade applications using Next.js, FastAPI, PostgreSQL, Docker, and Nginx across the full stack.
- Owned the design of scalable backend APIs, database schemas, and deployment architectures serving engineering, manufacturing, and quality control teams.
- Engineered and owned authentication infrastructure, including SSO, MFA, and zero-trust network access

via Cloudflare Tunnel.

- Identified and resolved slow database queries and API bottlenecks, reducing average response times by ~40% across high-traffic services.
- Drove cross-functional bug resolution with QA: traced issues to root cause, authored regression tests, and shipped fixes with zero disruption to end users.
- Sole engineer responsible for building and launching production workflow systems used daily by engineering, manufacturing, quality control, and management teams.

#### **Frontend Developer Intern**

Jan 2022 – Dec 2022

Skillship Foundation

Remote

- Owned development of responsive UI components across multiple product modules using React and Next.js, keeping the codebase modular, reusable, and extensible.
- Independently translated Figma mockups into pixel-accurate, cross-browser layouts using Tailwind CSS, ensuring consistent rendering across desktop, tablet, and mobile.
- Built and shipped reusable form components with client-side validation for registration, login, and course enrollment flows, reducing incomplete submissions.
- Owned REST API integration to dynamically fetch and render course listings, user progress, and event schedules, fully replacing previously hand-updated static pages.
- Took initiative to connect the frontend to an Oracle Database reporting endpoint, writing query calls and mapping returned data into structured dashboard components.
- Diagnosed and fixed cross-browser rendering inconsistencies in Safari and Firefox, tracing root causes to vendor-prefixed CSS and resolving them without brittle overrides.
- Contributed actively in sprint planning and design reviews, surfacing implementation blockers early and providing technical context that kept story estimates accurate before work began.

### **PROJECTS**

#### **Vyne**

#### **Software Engineer**

- Architected an AI-native correlation layer that unifies business and infrastructure events on a single timeline across all connected tools.
- Owned the core backend API using FastAPI and PostgreSQL, handling real-time event ingestion from multiple third-party platforms at scale.
- Set up and owned Docker containerization and GitHub Actions CI/CD pipelines, enforcing automated test gates on every push before production deployment.
- Designed and enforced RBAC and tenant-level data isolation across a shared multi-tenant infrastructure, ensuring strict security boundaries between customers.
- Led integrations with GitHub, Sentry, Datadog, HubSpot, Stripe, Linear, and PagerDuty, centralizing cross-team event data into a unified timeline.
- Delivered 15+ product modules, including chat, project management, docs, voice/video calls with live transcription, and AI-powered meeting recaps.

#### **NEXUS: Traveler Management and Workflow Automation**

#### **Software Engineer, American Circuits Inc.**

- Sole developer: designed and owned the schema, REST API, role hierarchy, and UI in parallel, then drove the full system from staging to live production.
- Designed and delivered a real-time ops dashboard for supervisors, surfacing job status, queue depths, operator activity, and QC sign-off tracking in a single view.
- Engineered QR code and barcode scanning to enforce step-by-step work-center routing across the production floor, eliminating manual check-in errors.
- Built Python automation for daily shift reporting and traveler throughput summaries, saving supervisors ~2 hours of manual work per day.
- Deployed and managed a cloud environment with health checks and uptime monitoring, supporting two

plants running concurrent production shifts.

- Owned compliance architecture: implemented RBAC, encrypted API calls, and a fully reconstructible audit trail on every action for manufacturing regulatory requirements.

### **Serenity Living**

### **Software Engineer**

- Sole engineer: took full ownership from brief to launch, delivering UI/UX design, backend services, and Vercel deployment end-to-end.
- Optimized Core Web Vitals through image compression, script deferral, and per-route code splitting, achieving a Lighthouse score of 95+ on mobile.
- Built and owned server-side logic for contact forms, newsletter signups, and booking requests via Next.js API routes, keeping all sensitive processing off the client.
- Implemented Open Graph tags, JSON-LD structured data, and an auto-generated sitemap to maximize search and social visibility.
- Shipped a fast, mobile-first frontend with a CMS-friendly content structure; zero engineering incidents since launch.
- Architected a CMS-driven content structure that empowers the client to add pages, swap images, and update copy independently without engineering involvement.

### **CERTIFICATIONS**

- AWS Solutions Architect, Associate -- O'Reilly
- Cisco CCNA Fundamentals -- O'Reilly
- SQL for IT Professionals -- O'Reilly
- Claude Code In Action -- Anthropic
- Google Data Analytics Professional Certificate -- Coursera